

SUMMER EXAMINATION 2026
Artificial Intelligence - Version A
BSIT - 3rd Semester Time: 90 Minutes Max Marks: 80

Name: _____ Roll No.: _____ Date: _____

1. An RNN is designed to process
(A) sequences (B) static tables only
(C) graphs only (D) images without order
2. If a fair binary event has probability 0.5, its odds of success are
(A) 1 to 1 (B) 1 to 2
(C) 2 to 1 (D) 0 to 1
3. Backtracking search usually assigns
(A) one variable at a time (B) all variables randomly
(C) only goal variables (D) no values
4. An LSTM uses gates to help preserve
(A) long-range dependencies (B) database keys
(C) pixel colors only (D) search depth
5. Forward chaining applies rules from
(A) known facts toward conclusions (B) goals toward facts
(C) random facts (D) only contradictions
6. Exploration in reinforcement learning means
(A) trying uncertain actions (B) always choosing the best known action
(C) removing rewards (D) stopping training
7. A utility-based agent uses utility to
(A) rank desirable outcomes (B) tokenize text
(C) prune syntax trees (D) count database rows
8. Gradient descent updates parameters using the
(A) negative loss gradient (B) positive labels only
(C) largest feature (D) random action
9. Data leakage occurs when training uses information
(A) unavailable at prediction time (B) from a larger batch
(C) with missing values (D) after normalization
10. Overfitting means a model fits training data well but
(A) generalizes poorly (B) has no parameters
(C) cannot train (D) has low variance always
11. A heuristic is admissible when it
(A) never overestimates true remaining cost (B) always equals true cost
(C) is negative (D) ignores the goal
12. An agent's sensors provide percepts, while its actuators produce
(A) actions (B) heuristics
(C) constraints (D) utilities
13. A model card is intended to document
(A) model purpose, limits, and performance (B) network routes
(C) database indexes (D) source code licenses only
14. A validation set is used mainly for
(A) model selection and tuning (B) final unbiased reporting
(C) data collection (D) encryption
15. Recall is the fraction of actual positives that are
(A) correctly found (B) predicted negative
(C) all negatives (D) true negatives only
16. The determinant of a 2 by 2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is
(A) $ad-bc$ (B) $ab+cd$
(C) $ac-bd$ (D) $a+b+c+d$
17. Arc consistency removes values that
(A) have no supporting neighbor value (B) are globally optimal
(C) are duplicated (D) are outside the root
18. A false positive occurs when a negative case is
(A) predicted positive (B) predicted negative
(C) left unlabeled (D) counted twice
19. The utility of an outcome represents its
(A) desirability (B) token count
(C) search depth (D) probability only
20. The Turing test evaluates whether a machine can
(A) converse indistinguishably from a human (B) sort numbers
(C) store facts (D) encrypt files
21. A learning rate that is too large may cause training to
(A) diverge (B) stop overfitting automatically
(C) produce labels (D) become deterministic
22. Q-learning estimates
(A) action values (B) class priors only
(C) parse probabilities (D) pixel edges
23. Simulated annealing sometimes accepts
(A) a worse move (B) only a goal move
(C) no move (D) a duplicate state
24. K-means assigns points to the nearest
(A) centroid (B) decision stump
(C) support vector (D) parse node
25. Minimax assumes that the opponent
(A) acts optimally (B) acts randomly
(C) never moves (D) maximizes our score
26. Alpha-beta pruning changes minimax results
(A) never (B) only at even depth
(C) always (D) only with chance nodes
27. For $TP=8$ and $FP=2$, precision is
(A) 0.80 (B) 0.20
(C) 0.60 (D) 0.90
28. A convolutional neural network is especially suited to
(A) spatial patterns (B) symbolic proofs only
(C) transaction logs only (D) shortest paths
29. Naive Bayes assumes features are conditionally independent given the
(A) class (B) learning rate
(C) action (D) cluster index
30. A vector dot product measures alignment and is largest for vectors pointing
(A) in similar directions (B) in opposite directions
(C) at random (D) with zero length only
31. A Bayesian network represents dependencies using a
(A) directed acyclic graph (B) stack
(C) confusion matrix (D) parse tree only
32. A model-based agent maintains
(A) an internal state (B) only a lookup table
(C) a random seed (D) a utility invoice
33. A confusion matrix summarizes
(A) classification outcomes (B) regression slopes
(C) search paths (D) cluster centroids
34. Weak AI is designed for
(A) a limited task (B) every intellectual task
(C) only robotics (D) only games
35. If every edge costs 1, a path of depth 4 has cost
(A) 4 (B) 1
(C) 0 (D) 8
36. Unsupervised learning seeks structure in
(A) unlabeled data (B) test labels
(C) utility functions (D) action policies
37. A constraint satisfaction problem consists of variables, domains, and
(A) constraints (B) rewards
(C) gradients (D) tokens
38. High bias commonly causes
(A) underfitting (B) memorization
(C) data leakage (D) high variance only
39. The MRV heuristic chooses the variable with
(A) fewest legal values (B) largest domain
(C) highest cost (D) most neighbors only
40. A fully observable environment provides the agent with
(A) complete relevant state information (B) only rewards
(C) no sensors (D) past actions only

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1. A reflex agent selects actions mainly from
(A) the current percept (B) a learned value function
(C) a planning graph (D) a proof tree
2. Differential privacy protects individuals by adding controlled
(A) noise (B) labels
(C) search depth (D) network hops
3. Iterative deepening combines the low memory of DFS with the completeness of
(A) BFS (B) hill climbing
(C) minimax (D) beam search
4. An episodic task environment has actions that
(A) do not affect later episodes (B) always require planning
(C) must be continuous (D) have no observations
5. A policy maps states to
(A) actions or action probabilities (B) class labels only
(C) database rows (D) heuristic costs
6. In alpha-beta pruning, alpha is the best value found for
(A) MAX so far (B) MIN so far
(C) the root heuristic (D) a random leaf
7. A ROC curve plots true positive rate against
(A) false positive rate (B) true negative rate
(C) loss (D) sample size
8. Responsible AI emphasizes fairness, transparency, privacy, and
(A) accountability (B) unlimited surveillance
(C) opaque decisions (D) forced automation
9. Breadth-first search expands the
(A) shallowest frontier node (B) lowest heuristic node
(C) deepest node (D) random node
10. Pooling in a CNN commonly reduces
(A) spatial dimensions (B) the number of classes
(C) training labels (D) token vocabulary only
11. AI is best described as the study of agents that perceive and act
(A) intelligent behavior (B) database normalization
(C) file compression (D) packet framing
12. A Markov decision process includes states, actions, transition model, rewards, and
(A) a discount factor (B) a compiler
(C) a parse tree (D) a domain vocabulary
13. A goal test checks whether a state satisfies
(A) the goal condition (B) the heuristic estimate
(C) the action cost (D) the percept history
14. Reinforcement learning learns from
(A) rewards and penalties (B) class labels only
(C) SQL joins (D) grammar rules
15. Knowledge engineering converts domain expertise into
(A) a knowledge base (B) a pixel matrix
(C) a routing table (D) a loss curve
16. An AI agent's performance measure specifies
(A) how success is evaluated (B) how files are compressed
(C) how tokens are split (D) how keys are stored
17. In STRIPS, an action has preconditions and
(A) effects (B) pixels
(C) gradients (D) priors
18. Fuzzy logic allows truth values
(A) between 0 and 1 (B) only -1 and 1
(C) only 0 or 1 (D) above 100
19. The F1 score is the harmonic mean of
(A) precision and recall (B) accuracy and loss
(C) bias and variance (D) speed and memory
20. Precision is the fraction of predicted positives that are
(A) actually positive (B) actually negative
(C) all observations (D) missed positives
21. A perceptron computes a weighted sum followed by
(A) an activation (B) a database join
(C) a graph search (D) a token split
22. Uniform-cost search selects the frontier path with
(A) lowest accumulated cost (B) lowest depth
(C) highest heuristic (D) most children
23. An expert system usually contains a knowledge base and an
(A) inference engine (B) image encoder
(C) order queue (D) optimizer
24. PCA projects data onto directions of greatest
(A) variance (B) class count
(C) reward (D) text frequency only
25. Cross-entropy penalizes a confident prediction that is
(A) wrong (B) correct
(C) uniform (D) missing a feature
26. A proposition is a statement that is
(A) true or false (B) always numerical
(C) a probability distribution (D) a neural layer
27. Forward planning starts at
(A) the initial state (B) the goal state
(C) the deepest leaf (D) the utility maximum
28. A learning rate that is too small usually makes convergence
(A) slow (B) impossible in every case
(C) random (D) independent of data
29. The discount factor gamma is normally constrained to the interval
(A) 0 to 1 (B) 1 to 2
(C) -2 to -1 (D) 10 to 100
30. In PEAS, the E stands for
(A) Environment (B) Execution
(C) Encoding (D) Evaluation
31. The Q-learning update uses the reward plus discounted
(A) next-state best Q value (B) current input size
(C) training accuracy (D) heuristic depth
32. Depth-first search uses a frontier organized as a
(A) stack (B) queue
(C) priority table (D) hash ring
33. The principal purpose of normalization in ML is to
(A) put features on comparable scales (B) create labels
(C) increase leakage (D) remove all outliers
34. A heuristic function estimates
(A) cost from a state to a goal (B) the exact training label
(C) network bandwidth (D) number of agents
35. Hill climbing can fail at a
(A) local maximum (B) root node only
(C) terminal compiler (D) balanced database
36. Backward planning works backward from
(A) the goal (B) the first percept
(C) the largest domain (D) the reward table
37. Backward chaining begins with
(A) a query goal (B) an empty database
(C) a sensor reading (D) a heuristic value
38. Supervised learning uses
(A) labeled examples (B) no observations
(C) only rewards (D) only rules
39. A rational agent chooses an action that maximizes
(A) expected performance (B) memory usage
(C) program length (D) randomness
40. A star search evaluates a node with
(A) $f(n)=g(n)+h(n)$ (B) $f(n)=g(n)-h(n)$
(C) $f(n)=h(n)/g(n)$ (D) $f(n)=g(n)h(n)$ only